Software Requirements Specification

for

Online Novel Reading and Writing System

Version 1.0 Approved

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Revision History

Name	Date	Reason For Changes
Version 1.0	30/01/2026	Initial Version
Version 2.0	10/2/2026	Error Correction

1.Introduction

1.1 Purpose

This project proposes the development of a web-based novel reading and writing system similar to Wattpad, which provides a digital space for both readers and writers to interact through literary content. The proposed system allows users to read novels across various genres, write and publish their own stories with multiple chapters, and interact with other users through features such as follows, and ratings. By providing an easy-to-use interface and organized content management, the system aims to encourage creativity, improve reading habits, and build an online community for literature enthusiasts.

This document presents the Software Requirements Specification (SRS) for the Online Novel Reading and Writing System. The purpose of this SRS document is to clearly define the functional and non-functional requirements of the system. It serves as a reference for developers, designers, testers, and stakeholders to understand the system's objectives, scope, features, and constraints, ensuring a common understanding of how the system should function throughout the development process.

1.2 Scope/ System Content

The Online Novel Reading and Writing Platform is designed to provide a web-based environment where users can register, authenticate, read novels, write and publish their own stories, and interact with other users. The system supports multiple genres, novel browsing and searching features, user authentication, content management, and administrative control.

Readers can explore stories, bookmark their favorite novels, adjust reading settings, and interact with content through follows, and ratings. Writers can create stories, upload and manage chapters, and edit or delete their own content, while editing privileges are restricted to the original author only. Administrators are responsible for managing users, monitoring and moderating content, and ensuring that the platform operates smoothly and securely.

The system aims to provide an engaging, user-friendly, and secure platform that promotes creative writing and convenient access to digital novels while demonstrating core Software Engineering concepts such as role-based access control, CRUD operations, and database-driven system design.

1.3 References

- **Books**
  Sommerville, I. *Software Engineering*, Pearson.
- **Websites**
  https://www.wattpad.com/home

  https://mmxianxia.com/

  https://www.measureevaluation.org/resources/publications/tl-19-34/at_download/document

1.4 Document Conventions

The document is made using Times New Roman where:
- Main-sections are indicated with bold letters font 18.
- Sub-sections also bolded with font 16.
- Body text is indicated with font 14.
- Functional requirement's description,input and output title are red.

An appendix A is added for acronyms and abbreviations

2. Overall Description

2.1 Product Functions

The subsection of the SRS provides a summary of the major functions that the Online Novel Reading and Writing System will perform. The key features of this system can be abstracted as follows:

● User-friendly web-based interface

● Online novel reading with chapter navigation and adjustable font size

● Online novel writing and publishing with multi-chapter support

● Novel browsing and search by title, genre, and status

● Genre-based novel categorization and display

● User registration, authentication, and profile management

● Bookmarking and personal writer library management

● User interaction features including follows, and ratings

● Role-based access for users(readers,writers) and administrators

● Administrative content and user moderation

● Data privacy and system security

2.2 User Classes and Characteristics

In the context of the Online Novel Reading and Writing System, the system targets three main user classes: readers, writers (authors), and administrators. A single user account may act as both a reader and a writer.

The Characteristics are:

Characteristics of Users (Readers):

- Readers can register, log in, and update their personal profile information.
- Readers can browse and read novels online.
- Readers can search novels by title, author name, or genre.
- Readers can view novel details including descriptions, chapters, ratings.
- Readers can bookmark novels for easy access.
- Readers can interact with novels by providing ratings.
- Readers can follow other users.

Characteristics of Writers (Authors):

- Authors can register and authenticate their accounts.
- Authors can create, edit, and delete their own stories and chapters.
- Authors can manage their published content through a personal dashboard.
- Authors can view reader interactions such as ratings on their stories.
- Authors can update their author profile information.

Characteristics of Administrators:

- Administrators can authenticate using role-based access control.
- Administrators can view and manage user accounts.
- Administrators can view and remove inappropriate stories or comments.
- Administrators can monitor system usage and basic platform statistics.
- Administrators can ensure compliance with platform rules and content policies.

2.3 Operating Environment

The Online Novel Reading and Writing System aims to provide a web-based platform accessible through modern web browsers including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge. The system is compatible with desktop computers, laptops, tablets, and mobile devices.

2.4 Design and Implementation Constraints

The system development is subject to the following constraints:

- The system must be developed using web technologies and supported frameworks.
- The database must securely store users' data and novel contents.
- The system must ensure secure handling of user data and intellectual property.
- Compliance with copyright laws and digital content distribution regulations is mandatory.
- The user interface must adhere to accessibility standards for a diverse reader base.

2.5 User Documentation

The user documentation for the Online Novel Reading and Writing System includes user manuals and online help guides.Documentation explains how to register, read novels, write and publish content, and manage user profiles.Admin documentation describes system management and content moderation procedures.

2.6 Assumptions and Dependencies

Assumptions and dependencies for the platform include:

- Users are responsible for maintaining account security and following platform guidelines.
- The system assumes users to have basic knowledge of web navigation and internet browsing.
- The system depends on server availability, database services, and hosting infrastructure.

3. External Interface Requirements

3.1 User Interfaces

The Online Novel Reading and Writing Platform provides a web-based interface accessible from modern web browsers. The user interface is designed to be intuitive, user-friendly, and responsive across different devices.

Reader/Writer Interfaces:

- Registration and login forms with username and password fields.
- Home page displaying all specific novels.
- Browse page displaying as a dropdown with genres so that users can search with genres.
- Search novels with their title name and users with @username options.
- Novel reading page with chapter navigation (Next / Previous) and adjustable font size.
- Story writing and chapter management dashboard for authors.
- Bookmark management page to view saved novels.
- User profile page showing personal information, written stories, and interaction history.
- Follow button is integrated into profile pages and rating buttons is integrated into novel pages.

Administration Interfaces:

- Admin login page with role-based authentication.
- User management dashboard to view, suspend, or delete user accounts.
- Content moderation panel to view, approve, or remove stories and comments.
- System statistics dashboard showing total users, stories, and interactions.

3.2 Hardware Interfaces

The system is designed to be platform-independent and requires minimal hardware resources. The following are the minimum hardware requirements:

- Desktop computer, laptop, tablet, or mobile device.
- Internet access (wired or wireless) with stable connection.
- Display with at least 1024x768 resolution for optimal viewing.
- Standard input devices such as keyboard and mouse or touchscreen.

3.3 Software Interfaces

The system interacts with the following software components:

- **Web Browsers:** Google Chrome, Mozilla Firefox, Safari, Microsoft Edge.
- **Backend Environment:** PHP for handling requests and server-side processing.
- **Database Management:** MySQL (or MariaDB) to store users, novels, chapters, bookmarks, follows, and ratings.
- **Frontend Technologies:** HTML, CSS, JavaScript, and Bootstrap for user interface and interactivity.
- **Development Tools:** VS Code for coding, phpMyAdmin for database management, XAMPP/WAMP as server environment.

3.4 Communication Interfaces

The system relies on internet-based communication for client-server interaction. All requests and responses between users and the server occur over standard HTTP/HTTPS protocols. Secure communication is enforced using HTTPS to protect user data during transmission.

4. System Features

4.1 Functional Requirements

Req – 1 : User Registration and Authentication

Description: The system shall allow new users to create an account and existing users to securely log in and log out.

Input: Registration requires username, email, and password; login requires username and password.

Output: The system validates the credentials against the MySQL database and establishes a secure user session upon success.

Req – 2 : Novel Browsing and Search

Description: The system must provide a central browsing page where users can view all available novels and search for specific titles.

Input: Users input search keywords or filter by genre and story status (Ongoing or Completed).

Output: The system displays a filtered list of novels matching the search criteria retrieved from the database.

Req – 3 : Reading System & Customization

Description: The system shall allow users to read individual chapters and customize the text display for a better user experience.

Input: User selects a chapter, navigates using "Next/Previous" buttons, and adjusts font size.

Output: The system displays the chapter content with the user's selected font settings applied.

Req – 4 : Bookmark Management

Description: The system shall allow readers to save their favorite novels to a personal section for quick access later.

Input: Clicking the "Bookmark" button on a specific novel's detail page.

Output: The system adds the novel to the user's dedicated bookmark section.

Req – 5 : Writing and Story Management (CRUD)

Description: Authors must be able to create new stories and perform CRUD (Create, Read, Update, Delete) operations on their chapters.

Input: Story metadata (title, description, genre) and text content for chapters.

Output: The system saves the data to the database and restricts editing permissions strictly to the original author.

Req – 6 : User Follow System

Description: The system shall enable social interaction between users through a follow mechanism.

Input: User clicks the "Follow" button on another user's public profile page.

Output: The system records the follow relationship in the database and updates the follower and following counts accordingly.

Req – 7 : Novel Rating System

Description: The system shall allow authenticated users to provide a rating for a novel in order to reflect their evaluation of the story.

Input: User selects a star rating (1–5 scale) on a novel's detail page.

Output: The system stores the rating in the database and updates the novel's overall average rating displayed on the platform.

Req – 8 : User Search and Public Profiles

Description: Users shall be able to find other members and view their public profiles and authored stories.

Input: Target username entered into the system's search bar.

Output: Display of the selected user's profile information and a list of their published works.

Req – 9 : Administrative Moderation

Description: The system shall provide a privileged role for administrators to manage users and moderate platform content.

Input: Admin credentials and selection of specific users or content to moderate.

Output: The system allows the Admin to suspend/ban users, delete stories.

5. Other Non-Functional Requirement

5.1 Performance Requirements

The system shall be designed to provide acceptable response times for all major functions. Pages such as the Bookshelf and novel listings should load within a reasonable time, ideally within three seconds under normal network conditions. Reading chapters, searching for novels, and filtering by genre should be performed efficiently without noticeable delay. The system should also be capable of handling multiple users accessing the platform simultaneously while maintaining stable performance.

5.2 Security Requirements

The system shall ensure that only authorized users can access protected features such as writing stories, managing chapters, and bookmarking novels. User authentication shall be required, and all user passwords shall be securely hashed before being stored in the database. The system shall restrict editing and deletion of novels and chapters to their original authors only. User inputs shall be validated to reduce the risk of common security threats such as SQL injection, and user sessions shall be securely managed to prevent unauthorized access.

5.3 Reliability Requirements

The system shall operate reliably during normal usage and ensure the consistency and accuracy of stored data. User-generated content such as novels, chapters, and bookmarks shall not be lost during normal operation. The database shall enforce integrity constraints to prevent invalid or inconsistent data. In the event of minor system failures, the system should continue functioning correctly, and critical data shall remain intact without corruption.

5.4 Usability Requirements

The system shall provide a simple, intuitive, and user-friendly interface that can be easily used by first-time users. Navigation between pages shall be clear and consistent throughout the application. The reading interface shall support adjustable font sizes to enhance reading comfort. User actions such as logging in, bookmarking novels, and saving chapters shall provide clear feedback so users understand the results of their interactions.

5.5 Maintainability and Support Requirements

The system shall be developed using a modular and well-structured design to simplify maintenance and future updates. The database schema and source code shall be clearly organized and documented to support debugging and enhancement. The system should allow bug fixes and minor improvements to be implemented without requiring major architectural changes. The design shall support future expansion, such as adding new features or improving existing functionality.

5.6 Privacy Requirements

The system shall protect user privacy by securely handling personal information such as email addresses and passwords. Sensitive data shall not be exposed to unauthorized users, and passwords shall never be stored in plain text. Public user profiles shall display only non-sensitive information, such as usernames and authored novels. The system shall not share user data with third parties and shall follow basic data protection principles.

5.7 Business Rules

A user shall register and log in before writing stories, bookmarking novels, rating novels, or following other users.

A single user account may act as both reader and writer.

Only the original author of a story may edit or delete that story and its chapters.

A story may contain multiple chapters.

A story shall belong to exactly one genre.

A story's status shall be either "Ongoing" or "Completed."

Users may bookmark multiple stories but cannot bookmark the same story more than once.

A user may follow multiple users but cannot follow the same user more than once.

A user may rate a story only once but may update their rating. The rating scale shall be from 1 to 5.

Administrators shall have privileges to manage users and stories, including suspending accounts and deleting inappropriate content to ensure compliance with platform policies.

6. Conclusion

6.1 Conclusion

In conclusion, the proposed Online Novel Reading and Writing Platform is designed to provide a user-friendly, secure, and engaging web-based environment for readers and writers. The system allows users to browse, read, rate novels, and follow other users, while providing writers the ability to create and manage stories and chapters. Administrators can monitor and moderate the platform to ensure proper usage and content quality.

This SRS document clearly defines the functional and non-functional requirements, system scope, user classes, interfaces, and business rules. It serves as a reference for developers, testers, and stakeholders to ensure that the system is implemented according to the defined objectives and standards. The completed system will demonstrate the application of Software Engineering principles, including modular design, database management, role-based access control, and user interaction modeling.

6.2 Appendix A: Glossary

DB: Database
SQL: Structured Query Language
Admin: Administrator
SRS: Software Requirements Specification
RAM: Random Access Memory
PC: Personal Computer
CPU: Central Processing Unit
OS: Operating System
Novel: A story with chapters
Chapter: A section of a novel
Bookmark: Saved novel for later reading
User: Person using the system
Writer: User who creates and manages stories
Reader: User who reads and interacts with stories